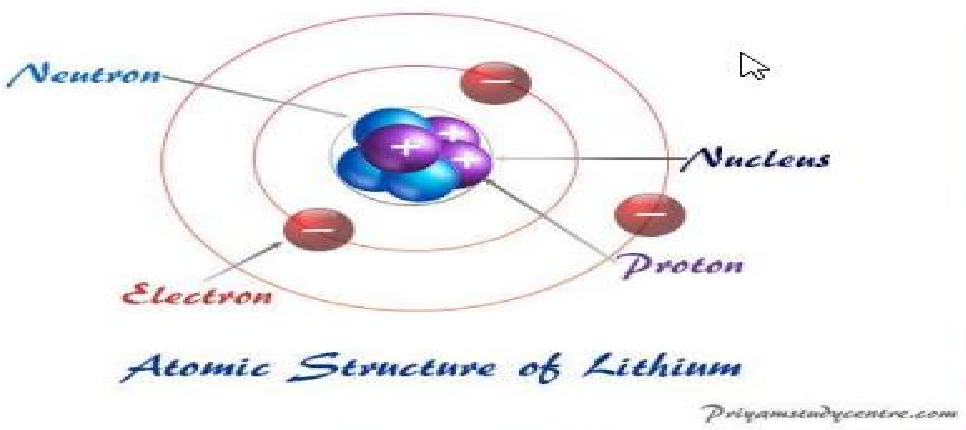
SS2 PHYSICS LESSON NOTES (THIRD TERM)
WEEK 4

1. HISTORICAL BACKGROUND OF THE ATOMIC MODEL

Ancient Ideas (Democritus, ~400 BC)

Proposed the idea that matter is composed of tiny indivisible particles called **"atomos"** (meaning uncuttable).
Had no experimental evidence — purely philosophical



2. DEVELOPMENT OF MODERN ATOMIC MODELS

2.1 Dalton's Atomic Model (1803)

Aspect	Details
Scientist	John Dalton
Nature of the Model	Indivisible solid sphere

Postulates:

All matter is composed of indivisible atoms
Atoms of the same element are identical in mass and properties.
Atoms combine in simple whole-number ratios to form compounds
Chemical reactions involve the rearrangement of atoms.

Strengths:

First scientific model supported by experimentation
Explains conservation of mass and definite proportions.

Limitations:

Did not include subatomic particles (electrons, protons, neutrons).
Failed to explain isotopes and chemical bonding.

2.2 Thomson's Model – "Plum Pudding Model" (1897)

Aspect	Details
Scientist	J.J. Thomson
Experiment	Cathode Ray Tube
Discovery	Electron

Model Description:

Atom is a positively charged sphere with negatively charged electrons embedded like "raisins in a pudding".

Key Observations from CRT Experiment:

Cathode rays (electrons) are negatively charged particles
They are deflected by electric and magnetic fields.

Strengths:

Introduced subatomic particles.
Explained electrical conductivity.

Limitations:

No central nucleus.
Could not explain atomic stability.
No explanation for spectral lines

2.3 Rutherford's Nuclear Model (1911)

Aspect	Details
Scientist	Ernest Rutherford
Experiment	Alpha Scattering (Gold Foil) Experiment

Setup: Alpha particles bombarded a thin gold foil.

Observations:

Most alpha particles passed through.
A few were deflected.
Some bounced back.

Conclusions

Atom is mostly empty space.

The atom has a small, dense, positively charged **nucleus**.
Electrons orbit the nucleus.

Strengths:

Introduced the nuclear structure of the atom.
Explained scattering of alpha particles.

Limitations:

Could not explain why negatively charged electrons don't spiral into the nucleus.
Could not explain atomic line spectra.

Bohr's Model (1913)

Scientist: Niels Bohr

Bohr improved Rutherford's atomic model by applying Quantum Theory.

Three Main Postulates:

Fixed Orbits: Electrons revolve in fixed circular paths called energy levels or shells.
Stable Energy: Electrons do not radiate energy while staying in these orbits.
Energy Transitions: Electrons emit or absorb energy only when jumping between different orbits.

Energy Level Formula:
$E_n = -13.6 / n^2$ eV
Where n = principal quantum number (1, 2, 3...)

Successes of Bohr's Model:

Explained the hydrogen atom's discrete spectral lines
Introduced quantization of angular momentum: $mvr = nh/2\pi$

Limitations of Bohr's Model:

Only worked accurately for hydrogen and single-electron atoms
Could not explain the Zeeman effect or Stark effect
Ignored the wave-particle duality of electro

Quantum Mechanical Model (1926–Present)

Scientists: Schrödinger, Heisenberg, de Broglie, Dirac

Key Ideas:

Electron Clouds: Electrons do not move in fixed orbits but exist in probability clouds called orbitals.
Wave Functions: Electron behavior is described using complex mathematical equations: $\psi(x,t)$

Heisenberg Uncertainty Principle: It is impossible to know both the exact position and momentum of an electron at the same time

de Broglie Wavelength: $\lambda = h/mv$ (Every particle has wave properties)

Strengths:

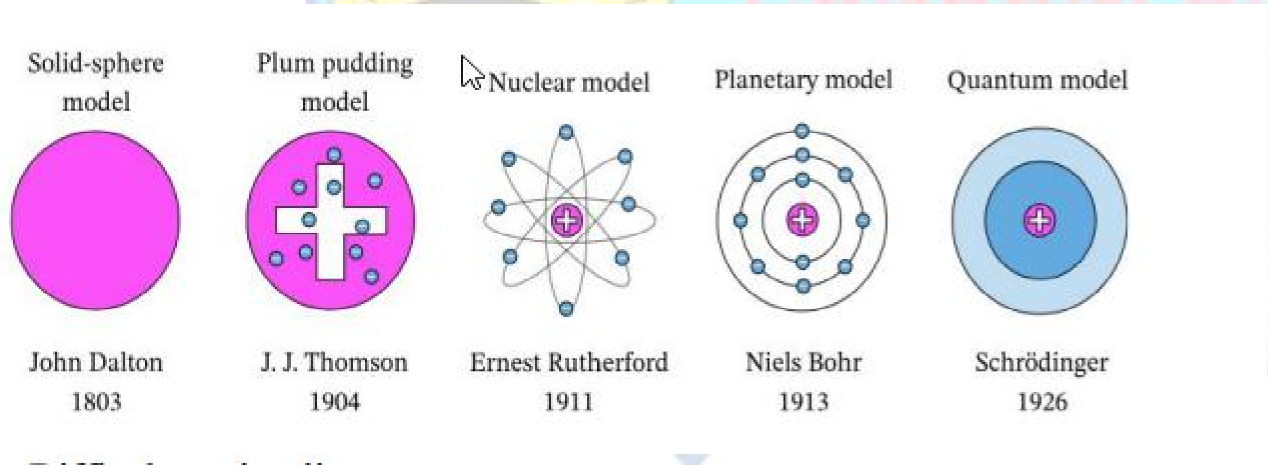
Describes all atoms, including complex multi-electron atoms

Foundation for quantum chemistry and particle physics

Limitations:

Highly mathematical and complex

Difficult to visualize physically



Comparison of Atomic Models

Model	Scientist	Description	Strengths	Limitations
Dalton	John Dalton	Solid indivisible atom	Explains conservation of mass	No subatomic structure
Thomson	J.J. Thomson	Electrons in positive sphere	Introduced electrons	No nucleus
Rutherford	Ernest Rutherford	Dense nucleus with orbiting electrons	Discovery of nucleus	Could not explain electron stability
Bohr	Niels Bohr	Quantized circular orbits	Explained hydrogen spectra	Inaccurate for complex atoms
Quantum	Schrödinger et al	Electron clouds and probability orbitals	Most accurate, includes wave nature	Highly mathematical

Applications of Atomic Models

Chemistry - Understanding molecular bonding and reactions
Spectroscopy - Explaining atomic spectra, emission and absorption lines
Electronics - Semiconductor design and operation
Medical Imaging - MRI and PET scans rely on atomic structure

Nuclear Physics - Nuclear fission, fusion, and atomic energy
Quantum Computing - Based on manipulation of electron states
Laser Technology - Based on electron energy level transitions

Evaluation Questions

Multiple Choice

1. Which model introduced the concept of quantized electron orbits?

A. Dalton's Model
B. Bohr's Model ✓
C. Rutherford's Model
D. Thomson's Mode

2. The discovery of the electron led to which atomic model?

A. Plum Pudding ✓
B. Nuclear Model
C. Planetary Model
D. Quantum Model

3. Which model introduced the idea of orbitals and probability clouds?

A. Bohr's Model
B. Dalton's Model
C. Quantum Mechanical Model ✓

D. Rutherford's Model

Theory Questions

List and explain four postulates of Dalton's atomic theory.
Describe Rutherford's gold foil experiment and state two of its conclusions.
Outline three limitations of Bohr's atomic model.
Compare Bohr's model with Schrödinger's quantum mechanical model.
Explain how atomic models helped in the development of X-ray and spectroscopy.

Assignment

(a) Draw and label diagrams for:

Bohr's hydrogen atom
Schrödinger's s and p orbitals
Rutherford's gold foil setup

(b) State how the concept of electron orbitals has influenced modern computer chip technology.

(c) Research the contribution of Heisenberg and how uncertainty plays a role in quantum mechanics.

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